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Hollow Boron-Doped Si/SiO$_x$ Nanospheres Embedded in VN/Nanopore-Assisted Carbon Conductive Network for Superior Lithium Storage

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KEYWORDS: Lithium-ion batteries, silicon, silicon oxide, vanadium nitride, graphitization carbon

ABSTRACT: SiO$_x$-based anode materials with high capacity and outstanding cycling performance have gained numerous attentions. Nevertheless, the poor electrical conductivity and non-negligible volume change hinder their further application in Li-ion batteries. Herein, we propose a new strategy to construct a hollow nanospheres with boron-doped Si/SiO$_x$ decorated by vanadium nitride nanoparticles as well as embedded in nitrogen-doped porous and partial graphitization carbon layer (B-Si/SiO$_x$@VN/PC). Benefiting from such structural and compositional features, the B-Si/SiO$_x$@VN/PC electrode exhibits a stable cycling capacity of 1237.1 mAh g^{-1} at a current density of 0.5 A g^{-1} with an appealing capacity retention of 87.0 % after 300 cycles. Additionally, it delivers high rate capabilities of 1139.4, 940.7 and 653.4 mAh g^{-1} at current densities of 2, 5 and 10 A g^{-1}, respectively and ranks among the best SiO$_x$-based anode materials.

The outstanding electrochemical performance can be ascribed to the reasons as following: (1) Its hollow structure makes the Li^+ transportation length decreased. (2) The existing nanopores facilitate the Li^+ insertion/desertion and accommodate the volume variation. (3) The nitrogen-doped partial graphitization carbon enhances the electrical conductivity and promotes the formation of stable solid electrolyte interphase (SEI) layers during repetitive Li^+ intercalation/extraction process.

1. INTRODUCTION

Currently, the increasing demand for high-energy-density lithium-ion batteries (LIBs) has stimulated much enthusiasm for developing advanced electrode materials with outstanding electrochemical performance.[1-3] Si-based anode materials have gained much attention because of their high theoretical specific capacities (Si: ~3579 mAh g^{-1}, SiO: ~2600 mAh g^{-1}, SiO_2: ~1680 mAh g^{-1}), low lithiation potential, environmental benignity and abundant resource.[4-8] Among them, elemental Si has the highest theoretical capacity but suffers from the large volume variation, which leads to the continuous capacity decay arising from the electrode pulverization.[2, 7] In contrast, Si/SiO_x, which is composed of active Si dispersing in less-active SiO_x matrix, exhibits improved cycling performance, despite showing lower specific capacities compared with pure silicon.[9-10] During the first lithiation process, Li_2O and Li_4SiO_4 will be *in situ* generated and can act as buffering components to accommodate a certain degree of volume change of Si.[4] Nevertheless, O-rich SiO_x with better cycling stability will sacrifice the reversible capacity and the initial coulombic efficiency (ICE). Meanwhile, the intrinsically low conductivity and nonnegligible volume change of SiO_x always lead to the poor rate performance and long-term cycling stability.

For the purpose of enhancing the conductivity of Si or SiO_x materials, various strategies have been developed. Typically, heteroatom (such as boron) doping, has been reported as a powerful strategy to enhance the electronic and ion conductivity of Si and SiO_x in LIBs.[11-13] Another effective way is surface modification/engineering by using highly conductive materials.[14] Besides the commonly carbonous materials, transition-metal nitrides (TMNs) have been widely used in LIBs as supporting materials to improve the electrical conductivity of other electrode materials due to their high electrical conductivity.[15] [16] Among TMNs, vanadium nitride (VN) shows many fascinating advantages such as high electronic conductivity (1.17×10^6 S cm^{-1}), high theoretical capacity (1043 mAh g^{-1}) and robust chemical stability.[15, 17-19] Unfortunately, the huge volume change of VN during the cycling process will deteriorate its cycling stability.

On the other hand, with respect to the pulverization issues of the electrode caused by the volume change, more efforts have focused on offering sufficient void to accommodate the expansion/shrinkage of active materials and constructing carbon protective layer to ensure the formation of stable SEI layer. [20-21]

Inspired by above considerations, in this work, we design and prepare a polyphasic nanocomposite consisting of hollow boron-doped Si/SiO_x, VN nanoparticles and carbon layer. The hollow structure with Si nanoparticles embedded in SiO_x matrix can alleviate the volume change to certain degree. Moreover, the introduction of VN/Ni can promote the formation of N-doped partial graphitization carbon that could enhance the electric conductivity of the composite and is more stable than the amorphous carbon shell during cycling. Meanwhile, the formed nanopores by acid treatment can not only accommodate the volume variation of VN and Si/SiO_x but also facilitate the Li^+ transportation.

2. EXPERIMENTAL SECTION

2.1 Preparation of hollow B-Si/SiO_x

Hollow silica nanospheres were synthesized by Stober's method mentioned in the previously reported literature.[22] A mixture of 1.0 g synthesized silica, 1.0 g magnesium powder, 10 g sodium chloride and 0.05 g boron oxide was placed into a tube furnace at 700 ℃ for 6 h under flowing forming gas (95% Ar with 5% H_2). After cooling down to the room temperature, the mixture was soaked in 1 M HCl solution for 6 h to remove the by-products. Then the sample was washed with deionized water (DI) and ethanol followed by vacuum drying to obtain B-Si/SiO_x. After immersing the B-Si/SiO_x in 10% HF solution for 1 h, the sample of B-Si could be prepared. The undoped Si/SiO_x was prepared under the identical condition without the addition of boron oxide.

2.2 Preparation of B-Si/SiO_x@VN/Ni and B-Si/SiO_x@VN

Typically, 0.1 g of B-Si/SiO_x was uniformly dispersed in methanol, followed by addition of 0.03 g of NH_4VO_3, 0.015 g of $NiCl_2·6H_2O$ and 0.02 g of oxalic acid ($H_2C_2O_4·2H_2O$) in sequence. After vigorous stirring for 30 min, H_2O_2 (1 mL) and concentrated HNO_3 solution (1 mL) were added dropwise to above solution for further stirring for 5 min. The mixture was transferred into a 100 mL autoclave and heated at 180 °C for 24 h. The product was obtained by filtration, ethanol washing and vacuum drying. Then, the dried powders were annealed at 600 °C for 2 h under ammonia atmosphere to prepare B-Si/SiO_x@VN/Ni. To prepare B-Si/SiO_x@VN, B-Si/SiO_x@VN/Ni sample was etched Ni by immersing into 0.1 M HCl solution and stirring at 30 °C for 1 h.

2.3 Preparation of B-Si/SiO_x@VN/PC

0.1 g B-Si/SiO_x@VN/Ni was dispersed into the mixed solution containing 120 mL of DI and 40 mL of ethanol under sonication for 1 h. Then 150 mg cetyltrimethylammonium bromide (CTAB) was dissolved in the suspension with continuous stirring for 20 min. After adding 32 mg of resorcinol, 46 μL of formaldehyde and 360 μL of ammonia aqueous solution (28 wt %), the mixture was magnetic stirred at 25 °C for 20 h. The obtained precipitates were washed and calcined at 900 ℃ for 4 h. Finally, B-Si/SiO_x@VN/PC composite was obtained by etching off the Ni. Meanwhile, B-Si/SiO_x@VN/C and B-Si/SiO_x@C samples were synthesized using the same experimental parameters except for using B-Si/SiO_x@VN and B-Si/SiO_x as the initial materials, respectively.

2.4 Materials characterization

The phase composition of the samples was determined by X-ray diffraction (XRD, DX-2600, Cu Kα radiation (λ = 1.54178 Å) that recorded the 2θ from 10°-80° with a scanning rate of 5° min^{-1}. The morphologies and structure details were characterized by scanning electron microscope (SEM, FEI Inspect F50 and JSM-5900LV) and transmission electron microscope (TEM, ZEISS Libra 200 FE). Raman spectra of the samples were recorded by a Raman microscope (532 nm, Renishaw Inxia). Inductively coupled plasma optical emission spectrometry (Agilent 730) was applied to certain the content of B element. X-ray photoelectron spectroscopy (XPS) analysis was performed on XSAM-800 spectrometer. The Brunauer-Emmett-Teller (BET) test was carried out with Micromeritics Instrument Corporation analyzer (ASAP 2020). For the microstructure observation of the cycling electrode, the batteries were disassembled and the obtained electrode materials were washed with dimethoxyethane (DME) followed by drying in an argon-filled glove box.

2.5 Electrochemical measurements

The electrochemical performances of all the as-prepared samples were evaluated by CR 2025 coin cells with lithium metal as the counter electrode. The working electrode was fabricated by a simple slurry process：homogenous slurries containing 70 % as-prepared samples, 15 % polyacrylic acid (PAA) and 15 % carbon black were uniformly pasted on the copper foil and vacuum-dried at 80 ℃ overnight. The loading of the active materials of each electrode was 1.4 ~ 1.6 mg cm^{-2}. 1 M $LiPF_6$ dissolved in a mixture of ethylene carbonate (EC) and diethylene carbonate (DEC) (1:1, v/v) with 5% fluoroethylene carbonate (FEC) was served as the electrolyte and Celgard 2400 polypropylene membrane as the separator. The galvanostatic discharge-charge experiments were performed in a potential range of 0.01 V to 3.0 V under different current densities using Land CT2001A system. All the specific capacities were calculated based on the mass of active materials instead of the silicon. Cyclic voltammetry (CV) measurements were conducted between 0.01-3.0 V with a scanning rate of 0.1 mV s^{-1} and electrochemical impedance spectroscopy (EIS) tests were performed by applying 5 mV AC amplitude perturbation from 100 kHz to 0.1 Hz on an electrochemical workstation (PARATAT 2273).

3. RESULTS AND DISCUSSION

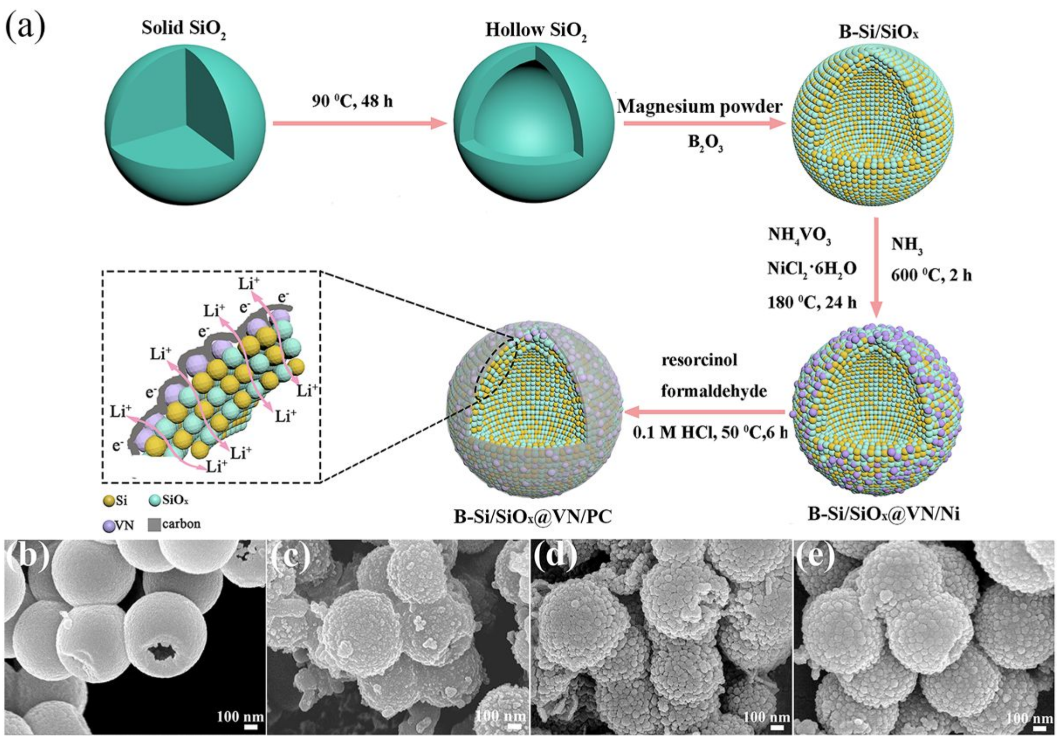

Figure 1. (a) Schematic illustration of preparation of B-Si/SiO_x@VN/PC composite, SEM images of (b) $HSiO_2$, (c) B-Si/SiO_x, (d) B-Si/SiO_x@VN/Ni and (e) B-Si/SiO_x@VN/PC.

The schematic preparation process of B-Si/SiO_x@VN/PC composite is presented in Figure 1(a). Firstly, the hollow SiO_2 nanospheres ($HSiO_2$) with ordered meso-aisles were synthesized by a spontaneous self-transformation method.[22-23] Then, the $HSiO_2$ and boric oxide were transferred into B-doped Si/SiO_x according to a modified magnesiothermic reduction process. For the B doping, the B atoms prefer to locate at the periphery of Si nanocrystals of the Si/SiO_x interface.[24-26] After a facile solvothermal approach and nitridation treatment, VN/Ni nanoparticles were uniformly anchored on the surface of B-Si/SiO_x. Then a thin layer of resorcinol–formaldehyde (RF) resin polymer coated the B-Si/SiO_x@VN/Ni by one-step sol-gel polymerization of resorcinol and formaldehyde in the presence of CTAB.[27] Finally, B-Si/SiO_x@VN/Ni was cladded by a very thin, porous and nitrogen-doped carbon layer after calcined in N_2 at 900℃ and further etched in HCl solution.

As shown in Figure 1 (b), the as-prepared $HSiO_2$ displays uniform spherical and hollow structures with a size range of 600-700 nm. The B-Si/SiO_x composite generated by modified magnesium reduction reaction basically retains the original morphology of $HSiO_2$ with the function of SiO_x matrix (Figure 1 (c)). The sample of B-doped Si cannot remain the spherical structure without the SiO_x matrix as shown in Figure S1. Figure 1 (d) shows the SEM image of B-Si/SiO_x@VN/Ni whose surfaces are decorated with nanoparticles, making the surfaces of B-Si/SiO_x rough. Additionally, as can be seen from Figure S2, the pure VN is constituted by agglomerated nanoparticles with size of 30-50 nm uniformly in the absence of B-Si/SiO_x matrix, probably causing the serious electrode pulverization during cycling process. After RF coating on B-Si/SiO_x@VN/Ni and an annealing process, the structure of B-Si/SiO_x@VN/PC nanospheres has not changed significantly except for relatively smooth surfaces as shown in Figure 1(e). Meanwhile, the morphologies of B-Si/SiO_x@VN/C and B-Si/SiO_x@C are characterized and shown in Figure S3.

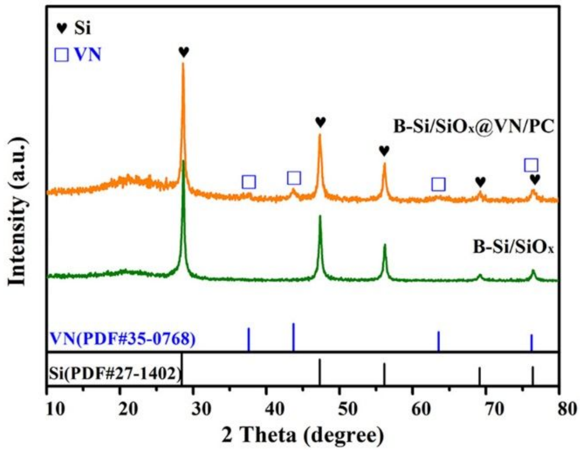

Figure 2. The XRD patterns of B-Si/SiO_x@VN/PC and B-Si/SiO_x.

Figure 2 displays the X-ray diffraction patterns of the B-Si/SiO_x@VN/PC and B-Si/SiO_x. In the pattern of B-Si/SiO_x, the diffraction peaks at 28.7°, 47.3°, 56.2°, 69.2° and 76.4° can be assigned to the (111), (220), (311), (400) and (311) planes of Si (JCPDS No.27-1402). Compared with un-doped Si/SiO_x (Figure S4), the peaks shift to higher angles, which can be ascribed to the decrease of lattice constant caused by the substitution of smaller B atoms for Si atoms.[28-29] The content of B element was 0.81 wt% based on ICP-OES result. The broad peak around 22° is arising from the presence of amorphous SiO_x.[14] The other peaks at 37.6°, 43.7°, 63.5° and 76.2° in B-Si/SiO_x@VN/PC are well matched with the (111), (200), (220) and (311) planes of VN

(JCPDS No.35-0768). Moreover, the XRD patterns of other as-prepared samples are shown in Figure S5 and Figure S6.

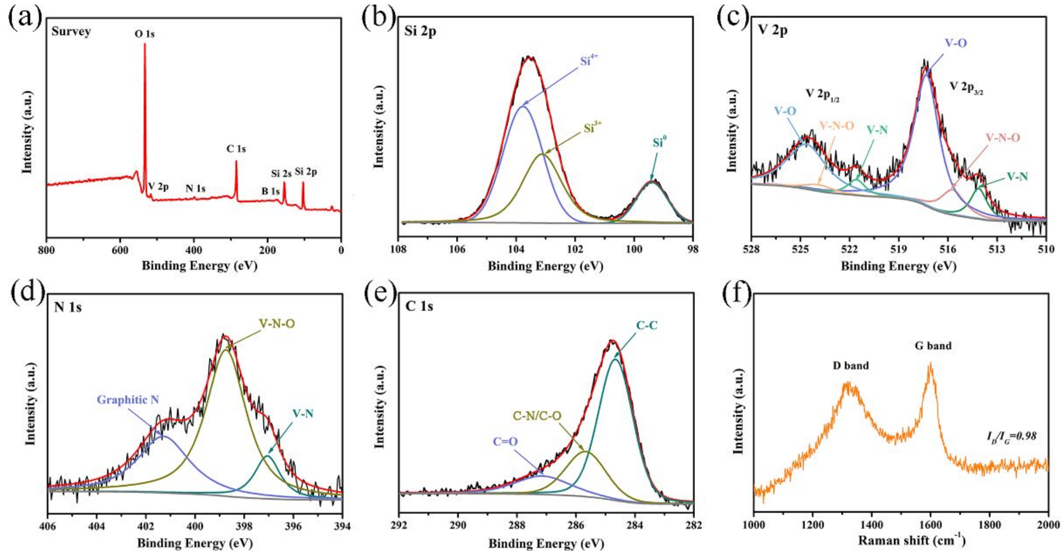

Figure 3. XPS survey spectrum of B-Si/SiO_x@VN/PC and the high-resolution spectra of (b) Si 2p, (c) V 2p, (d) N 1s and (e) C 1s, (f) Raman spectrum of B-Si/SiO_x@VN/PC.

The chemical composition and valence state of B-Si/SiO_x@VN/PC were further analyzed *via* XPS measurements. As shown in Figure 3(a), the full-survey-scan spectrum of B-Si/SiO_x@VN/PC displays the existence of Si, O, C, V, N and B. In the high-resolution spectrum of Si 2p (Figure 3 (b)), the peaks can be fitted into three peaks at 104.1 eV, 103.4 eV and 99.7 eV, corresponding to three types of Si species: Si^{4+}, Si^{3+} and Si^0.[30] The ratio of the three valence states was estimated from the area ratio of these peaks (Table S1). Then the O/Si atomic ratio in B-Si/SiO_x@VN/PC composite can be calculated to be 1.55 and the content of Si^0 in Si/SiO_x is calculated to be 7.33 wt% within the XPS probing depth.[31-32] Six deconvoluted peaks of V 2p can be found in Figure 3 (c). The peaks located at 514.1 and 521.6 eV are assigned to the V–N bond, while the other peaks are related to the vanadium oxides. The presence of oxide layers on VN is possibly caused by the oxidation during carbonization process of RF resins and the inevitable slight surface oxidation in the atmospheric conditions.[33-34] Remarkably, compared with the N 1s spectrum of B-Si/SiO_x@VN (Figure S7), the one of B-Si/SiO_x@VN/PC (Figure 3 (d)) displays a new peak relating to graphitic N (401.3 eV), [35-37] which could be ascribed to the fact that the loss of unstable N from B-Si/SiO_x@VN/Ni was possibly doped into carbon to form more stable graphitic N at high temperature.[38-39] The nitrogen doping into carbon will facilitate the ion penetration and accelerate the rate-limiting first electrons to transfer into the host materials.[39] The following deconvolution of C 1s shows the peaks situated at 284.6, 285.5 and 286.5 eV (Figure 3 (e)) corresponding to the C-C/C=C, C-N and C=O, respectively.[40-41] The contents of B, Si, O, V, N, C are 0.73 wt%, 39.08 wt%, 36.29 wt%, 8.31 wt%, 2.33 wt% and 13.27 wt%, respectively based on XPS analysis. Raman spectrum was further carried out to characterize the nature of carbon, and as shown in Figure 3(f) two obvious peaks were detected around 1329.1 and 1599.8 cm^{-1} tallying with the disorder-induced phonon mode (D-band) and graphitic band (G-band), respectively.[42] Furthermore, the I_D/I_G ratio in Raman spectrum (Figure 3(f)) is calculated to be 0.98, which is lower than that of B-Si/SiO_x@VN/C and B-Si/SiO_x@C (Figure S8), indicating the fact that a more efficient conductive network is constructed arising from the partial graphitization of amorphous carbon with the Ni catalyst.[30, 43-44,45] Moreover, a small red shift of the characteristic peak of Si in Figure S9 arises from the lattice deformation after B doping, which is in agreement with the XRD results.

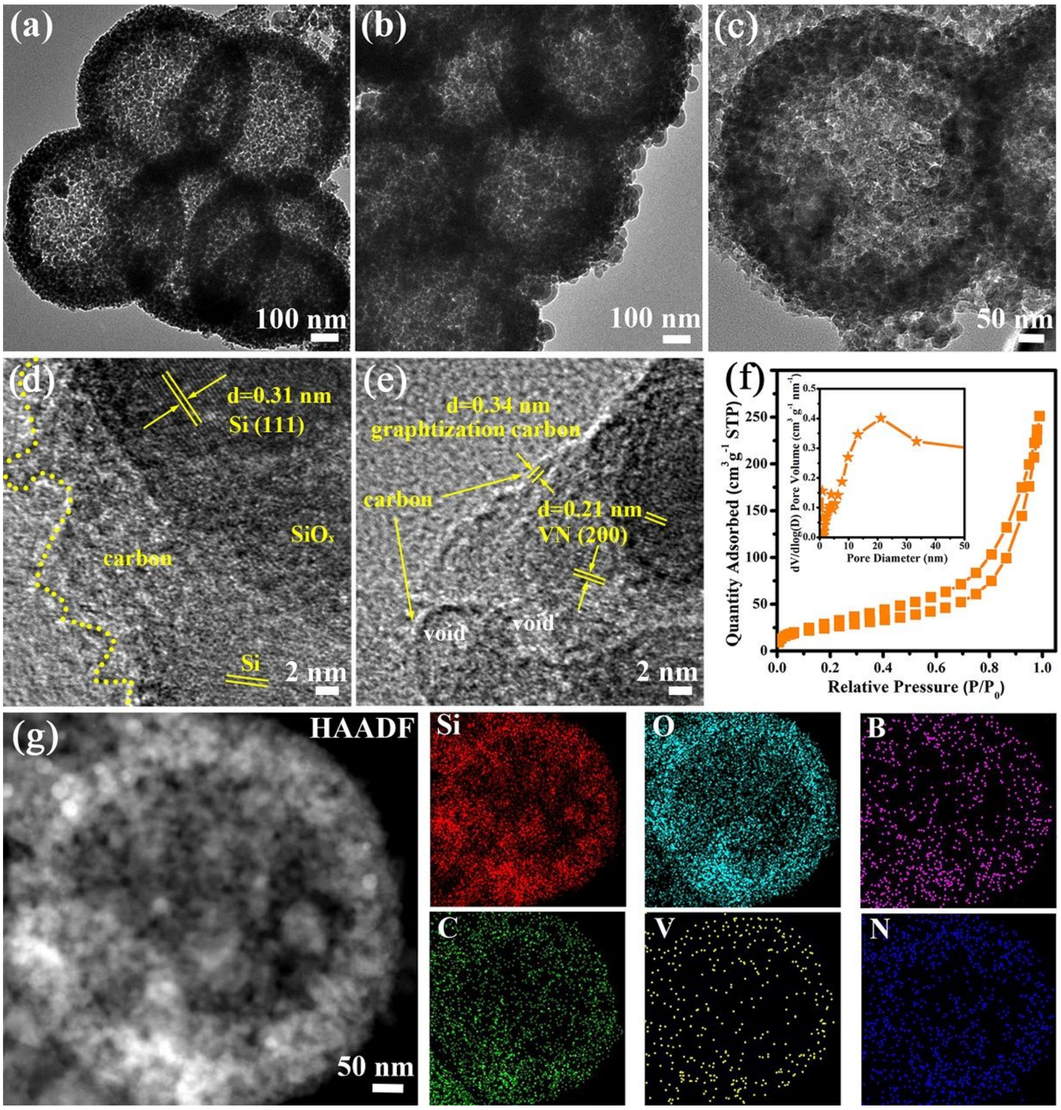

Figure 4. TEM images of (a) B-Si/SiO_x and (b) B-Si/SiO_x@VN/Ni, TEM (c) and HRTEM (d, e) images of B-Si/SiO_x@VN/PC in different regions, (f) N_2 absorption-desorption isotherms and corresponding pore-size distribution (inset) of B-Si/SiO_x@VN/PC, (g) HAADF-STEM image and the corresponding EDX elemental mappings of Si, O, B, C, V and N in a single Si/SiO_x@VN/PC nanosphere.

TEM was employed to make a further characterization for the structure of as-prepared samples. As can be seen in Figure 4 (a), B-Si/SiO_x exhibits a hollow structure and its shell composed of primary nanograins has an average thickness of about 90 nm. The small crystallites sizes (10-15 nm) are mainly ascribed to the mesoporous structure of hollow SiO_2 template that can prohibit the coarsening of the silicon crystallites during heat treatment.[46] The TEM image of B-Si/SiO_x@VN/Ni nanospheres (Figure 4 (b)) displays a rougher surface with observable nanoparticles randomly distributed. In addition, it can be found from Figure 4 (c) that the structure of B-Si/SiO_x@VN/PC nanospheres is highly porous compared to that of B-Si/SiO_x@VN/C (Figure S10), which is beneficial for the infiltration of electrolyte. The HRTEM image of B-Si/SiO_x@VN/PC (Figure 4 (d)) reveals the carbon coating layer and the embedment of Si nanograins in the amorphous SiO_x matrix. The interplanar spacing of 0.31 nm can be indexed to the (111) plane of the crystalline Si. As shown in Figure 4 (e), lattice fringe with the space of 0.21 nm is related to the (200) plane of VN and the nanoparticles are also wrapped by carbon layer. Interestingly, a thin and partial graphitization carbon as well as clear nanopores can be observed due to the presence of metallic Ni nanocatalyst. The porous structure of B-Si/SiO_x@VN/PC can also be confirmed by nitrogen adsorption/desorption measurement. Figure 4 (f) and Figure S11 display the representative type-IV isotherm profiles, suggesting the mesoporous structure of B-Si/SiO_x@VN/PC, B-Si/SiO_x@VN/C and B-Si/SiO_x. Compared with B-Si/SiO_x@VN/C, the increased pore volume (0.46 cm^3 g^{-1}) and average pore diameter

(19.5 nm) of B-Si/SiO_x@VN/PC are mainly induced by the nanopores which can release the volume effect of Si, SiO_x and VN during cycling. Meanwhile, the relatively small specific surface area of B-Si/SiO_x@VN/PC (177.8 m^2 g^{-1}) can be ascribed to the decrease of microporous structure because of the partial graphitization of amorphous carbon.[43, 47-48] Furthermore, the HAADF-STEM mapping images (Figure 4 (g)) indicate the good distribution of Si, O, B, C, V and N over B-Si/SiO_x@VN/PC nanospheres.

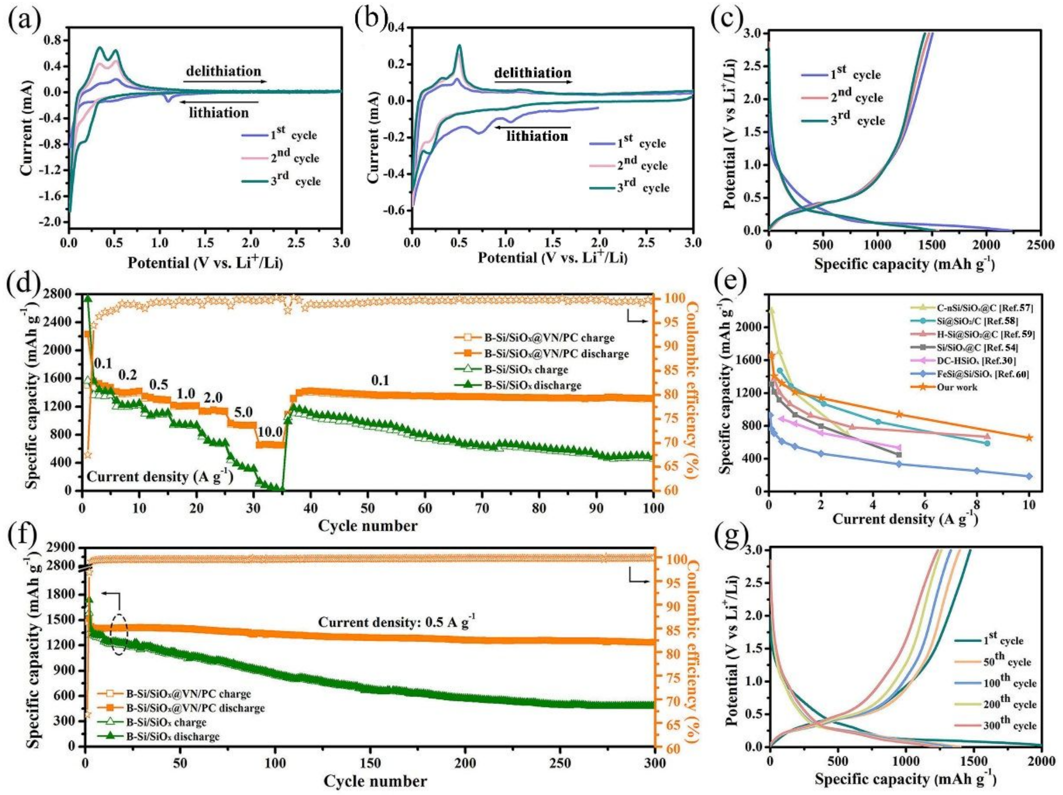

Figure 5. CV curves of (a) B-Si/SiO_x and (b) B-Si/SiO_x@VN/PC at a scanning rate of 0.1 mV s^{-1}, (c) charge-discharge curves of B-Si/SiO_x@VN/PC at 0.1 A g^{-1}, (d) rate performances of B-Si/SiO_x and B-Si/SiO_x@VN/PC electrodes at different current densities, (e) comparisons of the rate performance of B-Si/SiO_x@VN/PC with those of other reported similar anode materials, (f) cycling performances of B-Si/SiO_x and B-

Si/SiO$_x$@VN/PC electrodes at 0.1 A g^{-1} for the initial two cycles and 0.5 A g^{-1} for the following cycles, (g) charge-discharge curves of B-Si/SiO$_x$@VN/PC at different cycle numbers.

The lithium storage behaviors of the as-prepared samples were investigated in half-cells with a voltage window of 0.01–3.0 V. Figure 5 (a) and (b) show the CV curves of B-Si/SiO$_x$ and B-Si/SiO$_x$@VN/PC for the initial three cycles, respectively. During the first cathodic sweeping process, distinct reduction peaks at 1.0-1.3 V can be ascribed to the reductive decomposition of electrolyte additive FEC and the irreversible formation of solid electrolyte interface (SEI) layer on the surface of electrode.[49-50] Meanwhile, the peaks around 0.5-0.8 V are corresponding to the formation of the SEI layer in the conventional carbonate based electrolyte, the reduction of SiO$_x$ to Si, as well as the synchronous formation of Li_2O and a series of Li silicates[32, 51-52] The sharp cathodic peak closing to 0.01 V is considered to be the indication of the insertion of lithium ions into crystalline silicon.[6] In the following oxidation process, two peaks at ~0.32 and ~0.50 V are attributed to the extraction of lithium ions from Li$_x$Si and the formation of amorphous silicon, respectively.[53] After the first cycle, the reduction peak at ~0.18 V appears, which arises from the lithiation of amorphous silicon. Obviously, the peak becomes stronger and stronger with the activation process, indicating a gradual conversion from crystalline silicon to amorphous silicon.[54] In addition, a pair of broad anodic/cathodic peaks at about 0.96/1.17 V appear in Figure 5 (b), which is related to the Li insertion/extraction reaction of VN and has been commonly observed in transition-metal-nitride electrodes.[55-56] Figure 5 (c) shows the charge-discharge curves of B-Si/SiO_x@VN/PC for the first three cycles, and it can be found that the first discharge and charge capacities are 2229.3 and 1505.1 mAh g^{-1}, respectively with a coulombic efficiency of 66.8%. The large capacity loss is mainly due to the formation of SEI layer, Li_2O and Li-silicates. Fortunately, the coulombic efficiency increases to 94.5 % rapidly in the second cycle, which indicates that a stable SEI layer has been formed due to the partial graphitization carbon coating layer.

The rate capabilities of the as-prepared electrodes at various current densities are presented in Figure 5 (d) and Figure S12, and the corresponding data are listed in Table S2. Obviously, the B-Si/SiO_x@VN/PC exhibits a better high-rate capability than other samples, and it delivers a high discharge capacity of 653.4 mAh g^{-1} at 10 A g^{-1}. In addition, even after multiple rate cycles, the specific capacity of B-Si/SiO_x@VN/PC can almost recover the initial capacity and keep desirable stability in the following cycles when the current density returns to 0.1 A g^{-1}, indicating a high structural stability. As can be seen from Figure 5 (e), the rate performance of B-Si/SiO_x@VN/PC is also superior to those of many similar SiO_x-based materials in the literature.[30, 54, 57-60] In comparison, the bare B-Si/SiO_x electrode undergoes a sever capacity fading with the current densities increasing and its reversible capacity is only 61.6 mAh g^{-1} at a high current density of 10 A g^{-1} as shown in Figure 5 (d). The appealing rate performance of B-Si/SiO_x@VN/PC can be attributed to the hollow structure offering reduced Li^+ diffusion length and the existing nanopores facilitating the intercalation/deintercalation of Li^+. Besides, the nitrogen-doped partial graphitization carbon and distributed VN nanoparticles could serve as highly conductive framework to promote the electron charge transfer.

The cycling performance of the as-prepared samples were evaluated at 500 mA g^{-1} in a voltage window of 0.01-3.0 V. Compared with B-doped Si (Figure S13), the B-Si/SiO_x (Figure 5 (f)) exhibits better cycling stability caused by the *in situ* generated Li_2O and Li-silicates from the adjacent SiO_x matrix, which can accommodate the volume change and inhibit the electrochemical agglomeration of Si nanograins to a certain degree. However, it still suffers from continuous capacity decay caused by the pre-existing and subsequently generated Si nanoparticles on the surface of electrode, because they cannot be protected by the Li_2O and Li-silicates. To some extent, moderate carbon coating may serve as a buffering layer to prevent the pulverization of electrodes. Nevertheless, we still observe that continuous capacity loss in the as-prepared B-Si/SiO_x@C and B-Si/SiO_x@VN/C electrodes during cycling (Figure S14). Especially, the reversible capacity of B-Si/SiO_x@VN/C dramatically degrades after 176 cycles, which arises from the destruction of amorphous carbon layer during the long-term cycling. In contrast, the B-Si/SiO_x@VN/PC electrode shows an outstanding cycling performance with a high reversible capacity of 1237.1 mAh g^{-1} after 300 cycles, which indicates a capacity retention of 87.0% (Figure 4 (g)). It should be noted that the coulombic efficiency is more than 99.5% from the 10^{th} cycle and rise to 99.9% after the 300^{th} cycle. Additionally, the electrochemical performance of B-Si/SiO_x@VN/PC outperforms that of other reported SiO_x-based anode materials in LIBs (Table S3). The superior cycling stability can be ascribed to the reasons as follows: (1) The partial graphitic carbon shell coating on Si/SiO_x and VN nanoparticles is more stable than the amorphous carbon, which could prevent the continuous formation of fresh SEI layer on the electrode over cycling. (2) The decorated VN nanoparticles can make a little contribution to the capacity of the composite (Figure S15). (3) The suitable voids can relieve the stress and strain induced by volume variation of Si and VN nanoparticles during cycling upon Li^+ insertion/extraction.

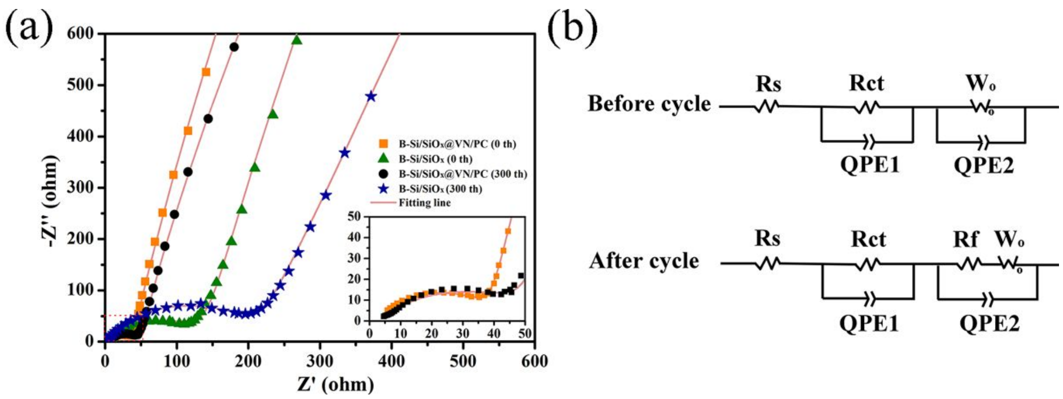

Figure 6. (a) Nyquist plots of B-Si/SiO_x and B-Si/SiO_x@VN/PC electrodes before cycling and after 300 cycles at a current density of 0.5 A g^{-1}, (b) the corresponding equivalent circuit model.

Electrochemical impedance spectroscopy (EIS) was conducted to further study the electrochemical reaction kinetics of the samples. All the Nyquist plots shown in Figure 6(a) and Figure S16 possess a conventional semicircle and a sloping straight line with the corresponding equivalent circuits presented in Figure 6 (b). The semicircle in high/medium frequency region can be attributed to the formation of SEI film (Rf) and charge transfer reaction at the interphase of electrodes and electrolytes (Rct), while the line in low frequency region is related to the ion diffusion behavior.[61] It is clear that the B-doped Si/SiO_x electrode has a lower Rct (139.6 Ω) than that of the undoped Si/SiO_x, which indicates that the boron doping is beneficial for the reversible redox reaction due to its offering effective electron conducting path for Si/SiO_x electrode.[11, 29] More importantly, the Rct of B-Si/SiO_x@VN/PC electrode (35.72 Ω) is the smallest among all the samples, which suggests that B-Si/SiO_x@VN/PC owns excellent electrical conductivity. After 300 cycles, the impedance of B-Si/SiO_x increased significantly because of the pulverization of the electrode caused by volume change. In contrast, the impedance of B-Si/SiO_x@VN/PC does not increase much, which indicates that a more stable structure can be retained during its cycling.

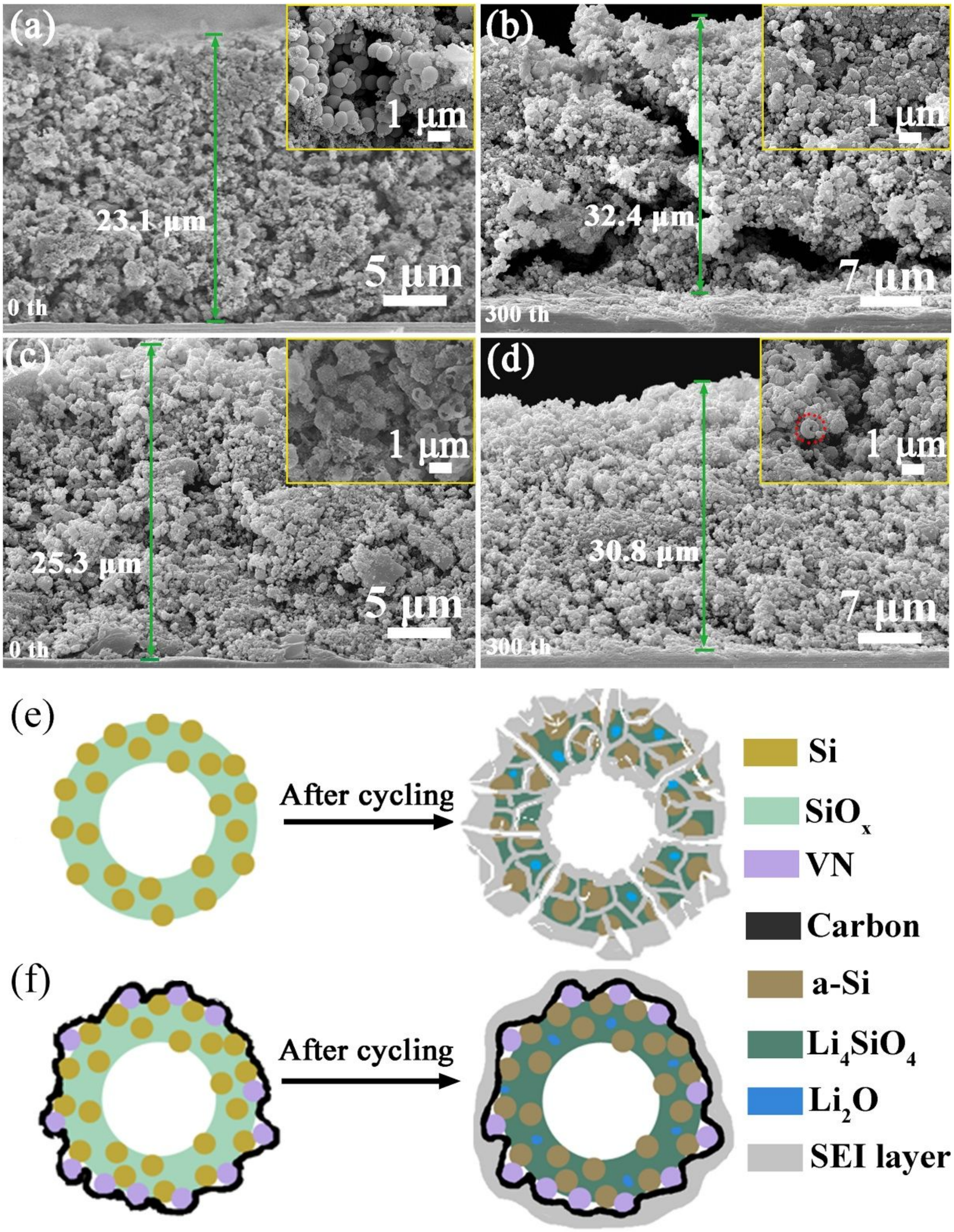

Figure 7. Cross-sectional SEM images of the electrodes before cycling and after 300 cycles at 0.5 A g^{-1}: (a) and (b) B-Si/SiO$_x$ electrode, (c) and (d) B-Si/SiO$_x$@VN/PC electrode, the inset shows the top-view of SEM images; The schematic diagram of (e) B-Si/SiO$_x$ and (f) B-Si/SiO$_x$@VN/PC before and after cycling.

To have a visual evidence for the stable structure of B-Si/SiO_x@VN/PC during cycling, FESEM characterization was employed to observe the difference between the morphology of B-Si/SiO_x and that of B-Si/SiO_x@VN/PC before cycling and after 300 cycles (Figure 7 (a)-(d)). Noticeably, the thickness of B-Si/SiO_x@VN/PC electrode increases by 21.7 % after cycling, which is smaller than that of B-Si/SiO_x (40.3 %), indicating that the volume variation of B-Si/SiO_x@VN/PC is suppressed. Moreover, distinct microcracks as well as the separation of electrode materials from current collector can be found in Figure 7 (b). This will cause a large charge transfer resistance of B-Si/SiO_x (Figure 6 (a)) after cycling as a result of the loss of electrical contact among active materials, their nearby particles and the current collector. On the contrary, similar disintegration does not happen in B-Si/SiO_x@VN/PC electrode and intact nanospheres can still be retained (Figure 7 (d)). Figure 7 (e) and (f) display the schematic diagram of B-Si/SiO_x and B-Si/SiO_x@VN/PC electrodes before and after cycling. Numerous SEI layer is coated directly on the surface of nanoparticles of B-Si/SiO_x nanospheres, which can be attributed to the repetitive volume expansion/contraction. Since the B-Si/SiO_x@VN was embedded in partial graphitization carbon shell with abundant internal nanopores, stable SEI layer was formed on the surface of every single B-Si/SiO_x@VN nanosphere to prevent the pulverization of electrode. As a result, B-Si/SiO_x@VN/PC shows outstanding electrochemical performance.

4. CONCLUSION

In summary, the hollow B-Si/SiO_x@VN/PC nanospheres have been prepared successfully by combining modified magnesiothermic reduction, solvothermal process and resorcinol formaldehyde (RF) coating. The hollow boron-doped Si/SiO_x nanospheres were decorated by vanadium nitride nanoparticles, and then partial graphitization carbon shell with nanopore embedded was coated on the surface of both VN and Si/SiO_x. Such an intriguing hollow nano-hybrid is beneficial for improving the rate capability and enhancing the cycling stability. Boron doping as well as the presence of partial graphitization carbon and VN nanoparticles can enhance the electronic conductivity of the composite. Moreover, the partial graphitization carbon shell can also prevent collapse of the hollow nanostructures and ensure the structural integrity during long-term cycling. It is believed that the structure design and the synthetic method in this work open a new avenue for other electrode materials with poor electrical conductivity and huge volume change during cycling.

ASSOCIATED CONTENT

Supporting Information.

SEM images of B-Si, VN, B-Si/SiO_x@VN/NC and B-Si/SiO_x@C; XRD patterns of B-Si/SiO_x, Si/SiO_x, precursor of B-Si/SiO_x@VN/Ni, B-Si/SiO_x@VN, B-Si/SiO_x@VN/Ni, B-Si/SiO_x@VN/C and B-Si/SiO_x@C; XPS spectra of B-Si/SiO_x@VN/PC and B-Si/SiO_x@VN; Raman spectra of B-Si/SiO_x@VN/C, B-Si/SiO_x@C, B-Si/SiO_x and Si/SiO_x; TEM image of B-Si/SiO_x@VN/C; N_2 absorption-desorption isotherms and pore-size distributions of B-Si/SiO_x and B-Si/SiO_x@VN/C. EDX spectrum of B-

Si/SiO_x@VN/PC; Electrochemical performances of B-Si, VN, B-Si/SiO_x@C, B-Si/SiO_x@VN and B-Si/SiO_x@VN/C; Nyquist plots of B-Si/SiO_x, B-Si/SiO_x@C and B-Si/SiO_x@VN/C electrodes; Abundance ratios of the different valance states of Si present in the B-Si/SiO_x@VN/PC; The discharge capacities at different current densities and ICE of all samples; Performance comparison of the present work with the reported SiO_x-based materials; The theoretical specific capacity of the composite.

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The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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